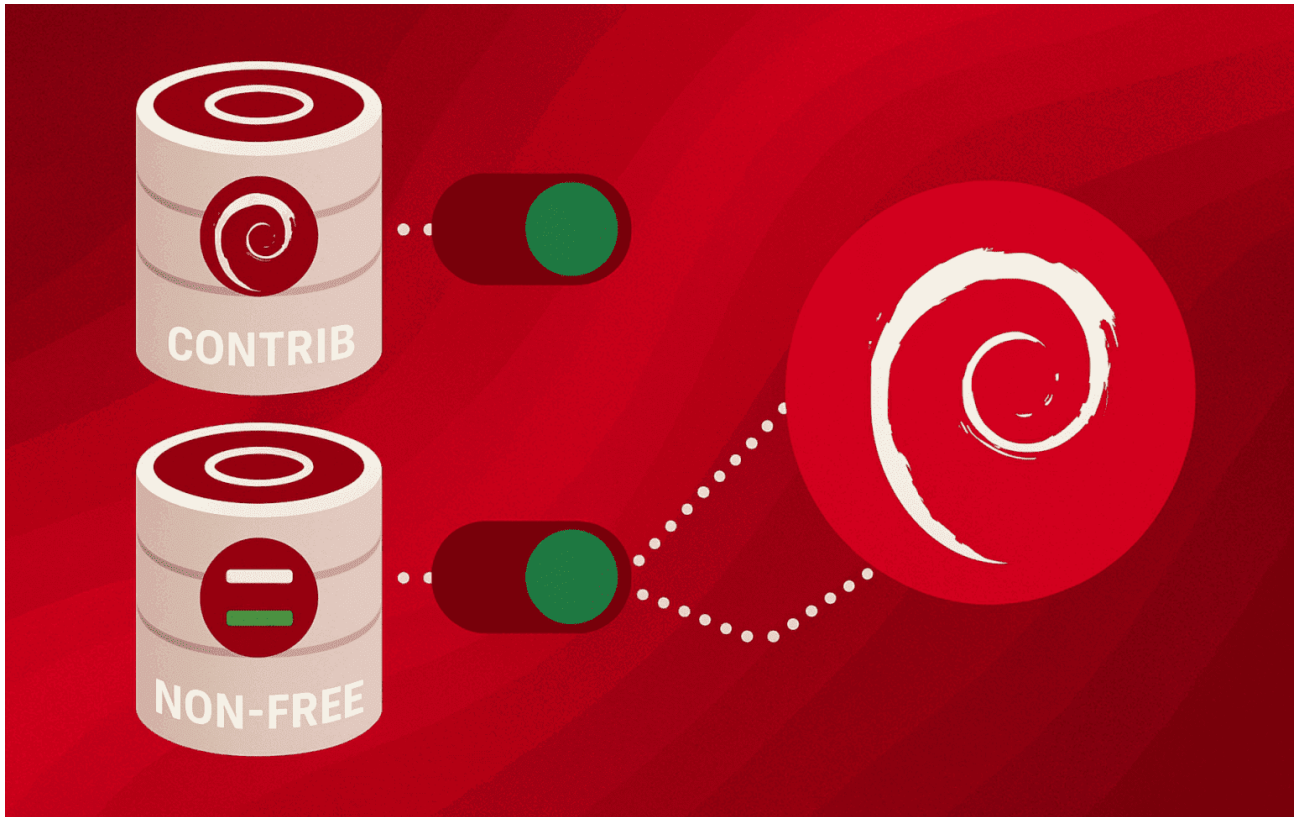


How to Enable Contrib and Non-Free on Debian 13, 12 and 11

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Debian keeps proprietary drivers, firmware, and some dependency wrappers outside the default main archive component, so APT can report packages such as `nvidia-driver`, `firmware-iwlwifi`, `intel-microcode`, or `unrar` as unavailable until the right components are enabled. To enable contrib and non-free repositories on Debian, update the Debian source entries for your release, then refresh APT so it can see those package indexes.

Debian 13 (trixie) and Debian 12 (bookworm) use `main contrib non-free non-free-firmware` when you want the full component set from Debian's own mirrors. Some Debian 12 and 13 installs may already include `non-free-firmware`, but `contrib` and `non-free` still need to be present for packages such as NVIDIA drivers, Steam dependencies, and `unrar`. Debian 11 (bullseye) uses `main contrib non-free` because `non-free-firmware` was introduced in Debian 12.

Enable Contrib and Non-Free Repos on Debian

Debian systems can store official archive entries in `DEB822` `.sources` files, legacy one-line `.list` files, or `/etc/apt/sources.list`. Inspect the active files first, then add components to the Debian entries that already exist. This preserves custom mirrors, source-package

entries, architectures, signing options, and suites instead of replacing the complete APT configuration.

Check Your Debian Release

Confirm the release codename before editing the component list:

```
grep '^VERSION_CODENAME=' /etc/os-release
```

Expected output shows one of the supported codenames:

```
VERSION_CODENAME=trixie
```

Use the Debian 13 component set for `trixie`, the Debian 12 component set for `bookworm`, or the Debian 11 component set for `bullseye`.

Inspect Active Debian Source Files

List active source definitions while keeping each DEB822 stanza grouped with its URI, suite, components, architecture, and signing option:

```
for file in /etc/apt/sources.list /etc/apt/sources.list.d/*.list
/etc/apt/sources.list.d/*.sources; do
    [ -f "$file" ] || continue
    if [[ "$file" == *.sources ]]; then
        awk -v file="$file" '
        BEGIN { RS=""; FS="\n"; shown=0 }
        {
            enabled=1
            for (i=1; i <= NF; i++)
                if (tolower($i) ~ /^enabled:[[:space:]]*no([[:space:]]|$)/)
                    enabled=0
            if (enabled) {
                if (!shown) { printf "\n### %s\n", file; shown=1 }
                printf "Stanza %d:\n", NR
                for (i=1; i <= NF; i++)
                    if ($i ~ /^(Types|URIs|Suites|Components|Architectures|Signed-
By|Enabled):/) print $i
                print ""
            }
        }' "$file"
    else
        entries=$(grep -nE '^[[:space:]]*(deb|deb-src)[[:space:]]' "$file" || true)
        if [ -n "$entries" ]; then
            printf '\n### %s\n%s\n' "$file" "$entries"
        fi
    fi
done
```

The command excludes DEB822 stanzas marked `Enabled: no`, commented legacy lines, and files with no active definitions. Fields such as `Types:`, `URIs:`, `Suites:`, and

Components: identify DEB822 files; lines beginning with deb or deb-src use the legacy format. A system can use both formats, and entries can share a URI and suite when they cover different source types or components. They become duplicates when they request the same index target, such as the same URI, suite, source type, component, and architecture.

Prepare APT for HTTPS Sources

Enabling components does not require changing the mirror protocol. However, the legacy examples use HTTPS. If you plan to change an existing HTTP entry to HTTPS, refresh the current package lists and make sure ca-certificates is installed before editing the URI:

```
sudo apt update
sudo apt install ca-certificates
```

If your current account cannot use sudo, set that up first with the [add a user to sudoers on Debian](#) guide.

Back Up the Debian Source File You Will Edit

Back up the exact active file shown by the inventory command. For the common DEB822 location, use:

```
sudo cp -a /etc/apt/sources.list.d/debian.sources "/root/debian.sources.$(date +%Y%m%d-%H%M%S).bak"
```

For the common legacy location, use:

```
sudo cp -a /etc/apt/sources.list "/root/sources.list.$(date +%Y%m%d-%H%M%S).bak"
```

Run only the command that matches the file you will edit. If the inventory shows a different filename, substitute that exact path and give the backup a matching name under /root. If you will edit both formats, back up both files. Keeping backups outside /etc/apt/sources.list.d/ prevents APT from treating them as source definitions.

Edit DEB822 Components on Debian 13 or 12

If the inventory shows Debian entries in a .sources file, open that exact file. The usual installer-created path is:

```
sudo nano /etc/apt/sources.list.d/debian.sources
```

For every official Debian stanza covering the stable, updates, or security suite, change only its Components: field to:

```
Components: main contrib non-free non-free-firmware
```

Edit DEB822 Components on Debian 11

For Debian 11 (bullseye), use this component field instead:

Components: main contrib non-free

Do not add non-free-firmware to Bullseye because that component does not exist on Debian 11. On every release, leave the existing Types:, URIs:, Suites:, Architectures:, and Signed-By: fields unchanged.

Edit Legacy sources.list Entries

If the inventory shows Debian entries as one-line definitions, open the listed sources.list or .list file. The common path is:

```
sudo nano /etc/apt/sources.list
```

Preserve each line's existing options, mirror URI, and suite, then append the release-appropriate components. Standard Debian 13 entries look like this:

```
deb https://deb.debian.org/debian trixie main contrib non-free non-free-firmware
deb https://deb.debian.org/debian trixie-updates main contrib non-free non-free-firmware
deb https://security.debian.org/debian-security trixie-security main contrib non-free non-free-firmware
```

For Debian 12, replace trixie with bookworm. For Debian 11, replace it with bullseye and end each line with main contrib non-free.

Check for Duplicate Debian Source Targets

Rerun the source inventory command before refreshing APT. Compare each active entry's source type, URI, suite, components, and architecture. If two entries request an overlapping index target, keep the intended definition and comment or disable only the overlap after backing up the affected files. Do not disable a complete source format merely because both .sources and .list files exist; they may own different source types, components, suites, or repositories.

Leave third-party entries such as Docker, Google, Microsoft, or application vendor repositories unchanged. Apply these component changes only to official Debian archive entries.

Refresh APT After Enabling Components

Refresh the package index so APT reads the new component list:

```
sudo apt update
```

A successful refresh should complete without duplicate-source, missing-component, or signature warnings.

Verify Debian Non-Free Repos Are Enabled

After `sudo apt update` finishes, confirm the package indexes that APT consumes and check package candidates. These checks give you stable proof without relying on volatile `apt update` mirror ordering.

Check Active Component Definitions

Ask APT which binary-package indexes it currently consumes. This view excludes disabled stanzas and keeps each component tied to its URI, suite, and architecture:

```
apt-get indextargets --format '$(IDENTIFIER) | $(SITE) | $(RELEASE) | $(COMPONENT)
| $(ARCHITECTURE)' \
    | grep '^Packages |' \
    | sort -u
```

For every official Debian 13 or 12 suite you edited, the separate Packages lines should collectively include:

```
main contrib non-free non-free-firmware
```

For every official Debian 11 suite you edited, the separate Packages lines should collectively include:

```
main contrib non-free
```

Confirm a Contrib Package Candidate

Check `ttf-mscorefonts-installer`, which is available from contrib on Debian 13, 12, and 11:

```
apt-cache policy ttf-mscorefonts-installer
```

Relevant Debian 13 output includes a candidate from trixie/contrib:

```
ttf-mscorefonts-installer:
  Installed: (none)
  Candidate: 3.8.1
  Version table:
     3.8.1 500
           500 https://deb.debian.org/debian trixie/contrib amd64 Packages
```

Version numbers can differ. Debian 12 and Debian 11 should return their candidates from bookworm/contrib and bullseye/contrib, respectively.

Confirm a Non-Free Package Candidate

Check a package that lives in non-free, such as the NVIDIA driver metapackage:

```
apt-cache policy nvidia-driver
```

Relevant Debian 13 output includes a candidate from the trixie/non-free component:

```
nvidia-driver:
  Installed: (none)
  Candidate: 550.163.01-2
  Version table:
    550.163.01-2 500
      500 https://deb.debian.org/debian trixie/non-free amd64 Packages
```

Version numbers differ by release. Debian 12 currently returns the driver from bookworm/non-free, and Debian 11 returns it from bullseye/non-free.

Confirm Non-Free Firmware on Debian 12 or 13

On Debian 12 or 13, check a firmware package to confirm non-free-firmware is visible:

```
apt-cache policy firmware-iwlwifi
```

Relevant Debian 13 output includes the trixie/non-free-firmware source line:

```
firmware-iwlwifi:
  Installed: (none)
  Candidate: 20250410-2
  Version table:
    20250410-2 500
      500 https://deb.debian.org/debian trixie/non-free-firmware amd64 Packages
```

On Debian 11, firmware-iwlwifi still comes from bullseye/non-free because Bullseye predates the split into non-free-firmware.

Understand Debian Archive Components

Debian separates packages by licensing and dependency rules. The [Debian archive policy](#) defines the official component boundaries, while the [2022 Debian vote on non-free firmware](#) explains why firmware moved into its own component starting with Debian 12.

Component	What It Contains	Common Reason to Enable It
main	DFSG-free software that does not require packages outside main to build or run.	Default Debian system packages, desktop tools, servers, libraries, and development tools.
contrib	DFSG-free software that depends on packages outside main to build or run.	Wrappers, helpers, and packages that need a non-free runtime or companion package, such as some virtualization workflows covered in install VirtualBox on Debian .
non-free	Software that does not meet the Debian Free Software Guidelines.	Proprietary drivers, restricted utilities, and packages such as install unrar on Debian .
non-free-firmware	Device firmware separated from broader non-free software in Debian 12 and newer.	WiFi firmware, GPU firmware, CPU microcode, and other hardware initialization packages.

Enabling these components does not install proprietary software by itself. APT downloads metadata for the extra components, then installs packages only when you explicitly run an install command.

Enable Only the Components You Need

The DEB822 and legacy instructions enable the common full set for each supported Debian release. If you only need firmware on Debian 12 or 13, you can use `main non-free-firmware` instead. If you need NVIDIA drivers, Steam dependencies, unrar, or packages that depend on non-free software, include `contrib` and `non-free` too.

Keep Backports Separate

The source definitions here do not enable `trixie-backports` or `bookworm-backports`. Backports is a separate suite for newer packages on stable releases, not a required part of enabling `contrib`, `non-free`, or `non-free-firmware`. Use the [install backports and experimental repositories on Debian](#) guide if you need that workflow.

Debian 11 (bullseye) LTS runs until August 31, 2026 for LTS-supported architectures, but Bullseye no longer has active backports and does not support the `non-free-firmware` component. Keep Bullseye source entries limited to `main contrib non-free`, or plan an [upgrade from Debian 11 to Debian 12](#) if you need the newer component layout.

Troubleshoot Debian Non-Free Repository Issues

Most failures come from duplicate source entries, using a Debian 12 or 13 component list on Debian 11, missing HTTPS certificates, or an incorrect signing-key configuration.

APT Reports Duplicate Repository Entries

Duplicate warnings mean more than one active definition requests the same Debian index target:

```
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in
/etc/apt/sources.list:1 and /etc/apt/sources.list.d/debian.sources:1
```

Rerun the stanza-aware source inventory from the first section so active legacy entries and grouped DEB822 fields are visible without comments or disabled stanzas.

Compare the source type (deb or deb-src), Debian URI, suite, components, and architecture in each file. Back up the files involved, keep the intended definition, and comment, disable, or narrow only the entry that overlaps the same index target. Different source formats can remain active when they own different types, components, suites, architectures, or repositories.

If APT instead reports `Conflicting values set for option Signed-By`, compare the `Signed-By:` field or bracketed `signed-by=` option. Entries for the same URI and suite must agree on source-wide options such as `Signed-By`, even when their components do not overlap.

After correcting the exact duplicate, refresh APT:

```
sudo apt update
```

Debian 11 Reports Missing non-free-firmware

Debian 11 prints a component warning if `non-free-firmware` appears in a Bullseye source entry:

```
W: Skipping acquire of configured file 'non-free-firmware/binary-amd64/Packages' as
repository 'https://deb.debian.org/debian bullseye InRelease' doesn't have the
component 'non-free-firmware' (component misspelt in sources.list?)
```

Rerun the stanza-aware source inventory from the first section. Because it excludes comments and `Enabled: no stanzas`, any Bullseye definition that still lists `non-free-firmware` is active. Edit that exact file, remove `non-free-firmware` from the Bullseye entry, then run:

```
sudo apt update
```

APT Cannot Find Firmware or Microcode Packages

If a firmware or microcode package is missing, APT may show a package-candidate error like this:

```
E: Package 'intel-microcode' has no installation candidate
```

The related message ensure you enable the 'non-free-firmware' component in the apt sources means APT cannot see the firmware index that contains packages such as intel-microcode, amd64-microcode, and firmware-iwlwifi. On Debian 12 or 13, rerun the apt-get indextargets check under **Check Active Component Definitions**. If the official Debian package-index lines do not include non-free-firmware, add it to each affected active definition and run `sudo apt update`. On Debian 11, use non-free instead because firmware packages remain in that component.

APT Fails with TLS Certificate Errors

HTTPS source entries require the ca-certificates package. If APT reports an untrusted issuer, an unknown certificate authority, or another certificate-verification failure on a minimal or netinstall system, the package may be missing.

If your previous HTTP-based sources still work, install the certificate package first:

```
sudo apt install ca-certificates
```

Then rerun the update:

```
sudo apt update
```

A connection timeout, connection refusal, DNS failure, or generic port 443 error has a different cause. Check the network, DNS resolver, proxy, and firewall path instead of reinstalling certificates for those symptoms.

APT Fails with NO_PUBKEY

A NO_PUBKEY warning identifies a repository whose metadata APT cannot verify. Read the failing URI in the apt update output first. If it belongs to a third-party vendor, repair that vendor's key configuration instead of changing Debian's archive keyring. Continue here only when the failing URI is an official Debian archive such as `deb.debian.org` or `security.debian.org`.

Rerun the source inventory and inspect the affected Debian entry's Signed-By: field or bracketed signed-by= option. An omitted option can be valid because APT then uses its configured trusted keyrings; do not add a per-source path merely because the option is absent. If the entry names the common Debian keyring path, verify that the file exists and is owned by the archive-keyring package:

```
ls -l /usr/share/keyrings/debian-archive-keyring.gpg
dpkg -S /usr/share/keyrings/debian-archive-keyring.gpg
```

If a configured Signed-By path is missing or incorrect, restore its correct value from your backup or another equivalent working official Debian entry. Reinstalling the keyring alone does not correct a mistyped source option. If the source uses the default trusted keyrings or its configured path is correct, check whether the package is installed:

```
dpkg -l debian-archive-keyring | grep '^ii'
```

If the package is missing or damaged and another official Debian source still verifies correctly, reinstall it from that working source:

```
sudo apt install --reinstall debian-archive-keyring
sudo apt update
```

Do not bypass signature checks with `trusted=yes` or insecure APT options. If no configured Debian source can be verified, recover the keyring through trusted Debian installation media or another separately verified Debian package source before refreshing APT.

software-properties-common Is Not Needed

Debian does not need Ubuntu's `add-apt-repository` workflow to enable official Debian archive components. You do not need `software-properties-common` for these DEB822 or legacy source edits.

Disable Contrib and Non-Free Repos on Debian

To disable components later, rerun the source inventory command and edit the exact Debian definitions you changed. Restore each component list to its pre-edit value; do not automatically reduce every entry to `main`. Debian 12 and 13 installers often enable `non-free-firmware` by default, so removing it when it was already present can stop firmware updates.

The following commands show the common paths. Substitute the exact filename reported by the inventory command. For the DEB822 format, open:

```
sudo nano /etc/apt/sources.list.d/debian.sources
```

For the legacy format, open `sources.list` instead:

```
sudo nano /etc/apt/sources.list
```

On Debian 12 or 13, a common pre-edit value is `Components: main non-free-firmware`. If that was your baseline, change:

```
Components: main contrib non-free non-free-firmware
```

Back to:

Components: main non-free-firmware

If the Debian 12 or 13 entry originally contained only main, restore Components: main instead. For Debian 11, a common reversal is Components: main contrib non-free back to Components: main. Use the backup or your recorded pre-edit value rather than assuming either baseline.

If you use legacy sources.list lines, apply the same rule: remove only the component names that this guide added. Preserve any component that existed beforehand, along with the line's options, URI, and suite. Refresh APT after saving the file:

```
sudo apt update
```

If you created a backup earlier, restore the complete file only when no other source changes have been made since that backup. Otherwise, reverse the component edit manually so newer unrelated changes remain intact.

Removing a component prevents future installs and updates from that component. Already installed packages remain on the system, but firmware, driver, or application packages from a removed component may stop receiving updates.

Conclusion

APT can now resolve packages from contrib and non-free, plus non-free-firmware on Debian 12 or 13, without replacing custom mirrors or signing options. From here, you can [install NVIDIA drivers on Debian](#) or [install Steam on Debian](#). For an optional helper that manages separate third-party repositories, see how to [manage third-party repos with extrepo on Debian](#).